

NPTEL Course on
"Computer System Optimization for ML"

Lecture #18

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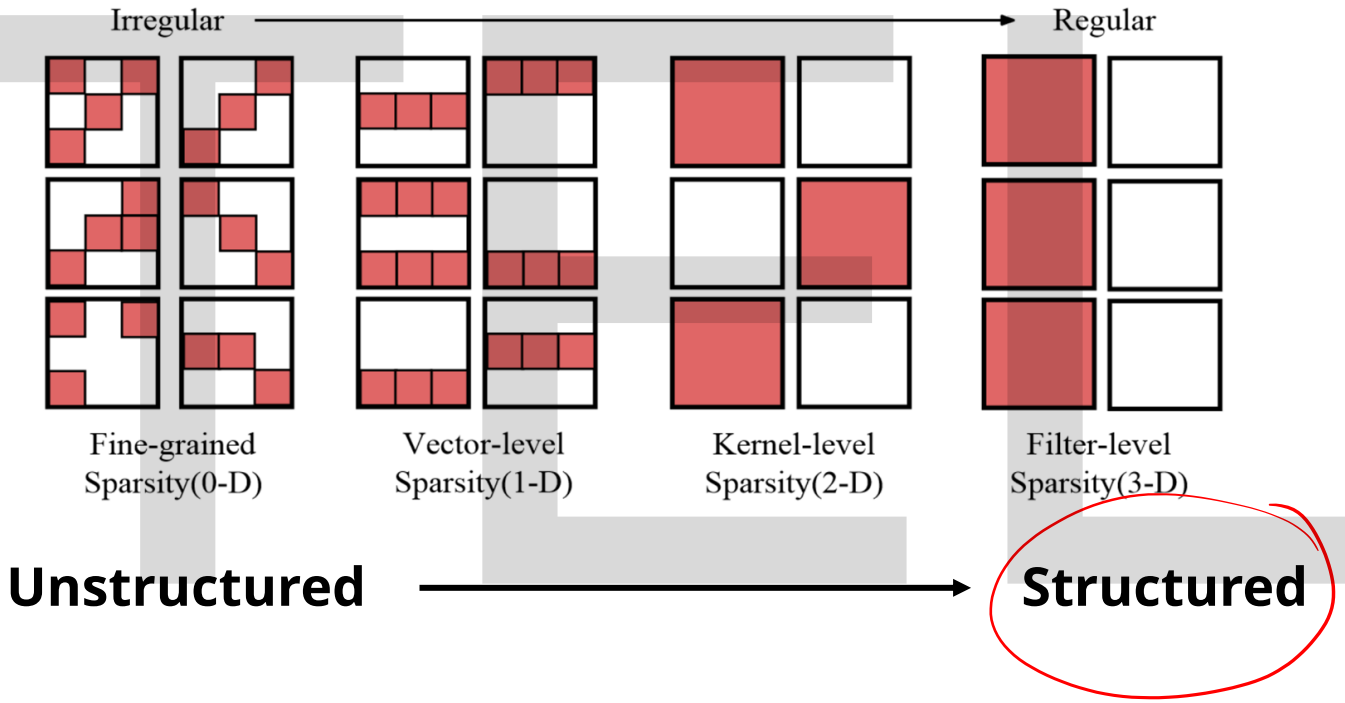
Agenda

- Recap
- DNN Pruning (contd.)

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Pruning Granularity in DNNs (2)

- **Unstructured**
  - Finer control over selecting elements to prune
  - Can provide better sparsity to accuracy ratio, i.e., smaller network with higher accuracy
  - Difficult to implement
- **Structured**
  - Less control over selecting elements to prune
  - Relatively less sparsity to accuracy ratio
  - Easier to implement



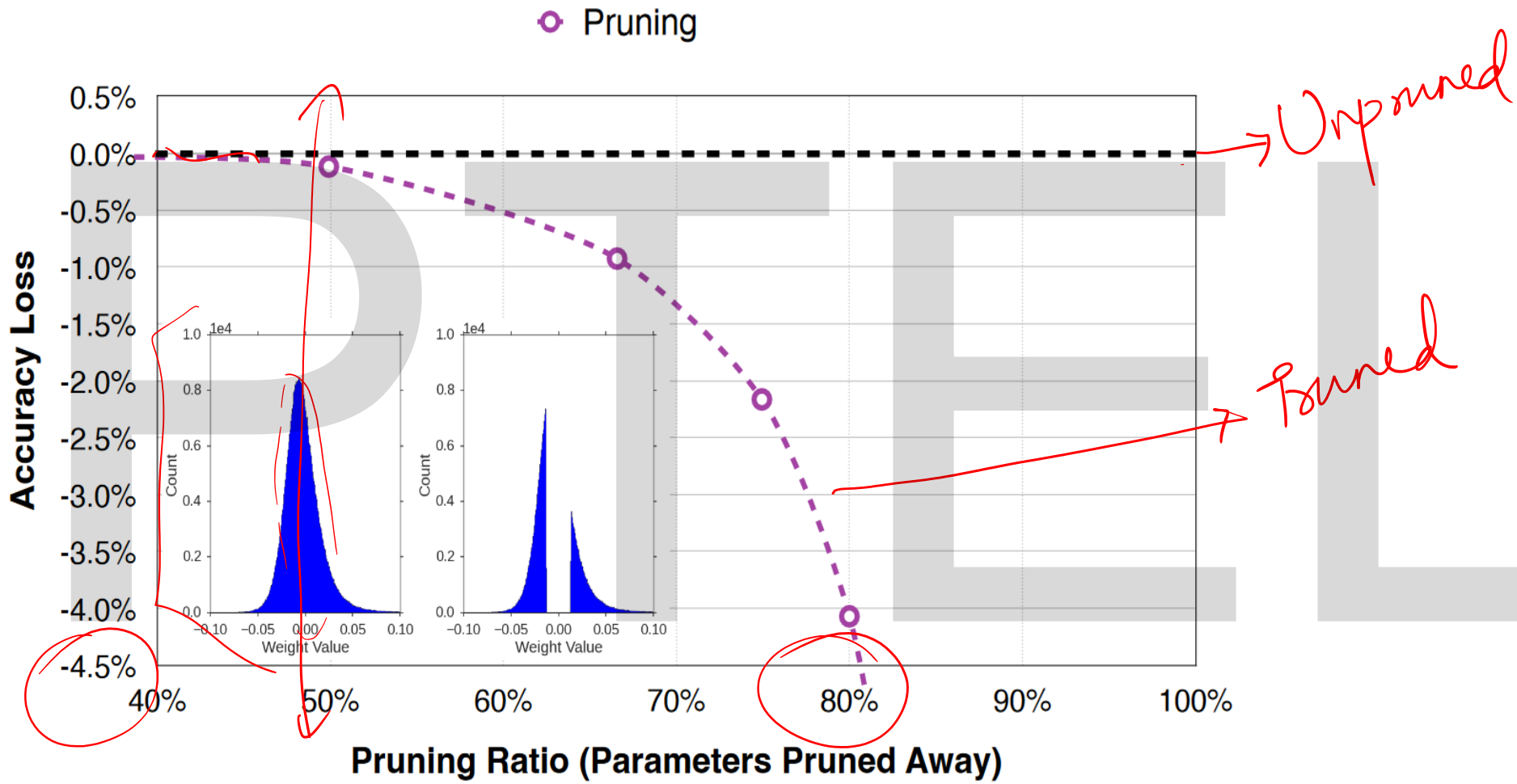
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Agenda

- Recap
- **DNN Pruning (contd.)**

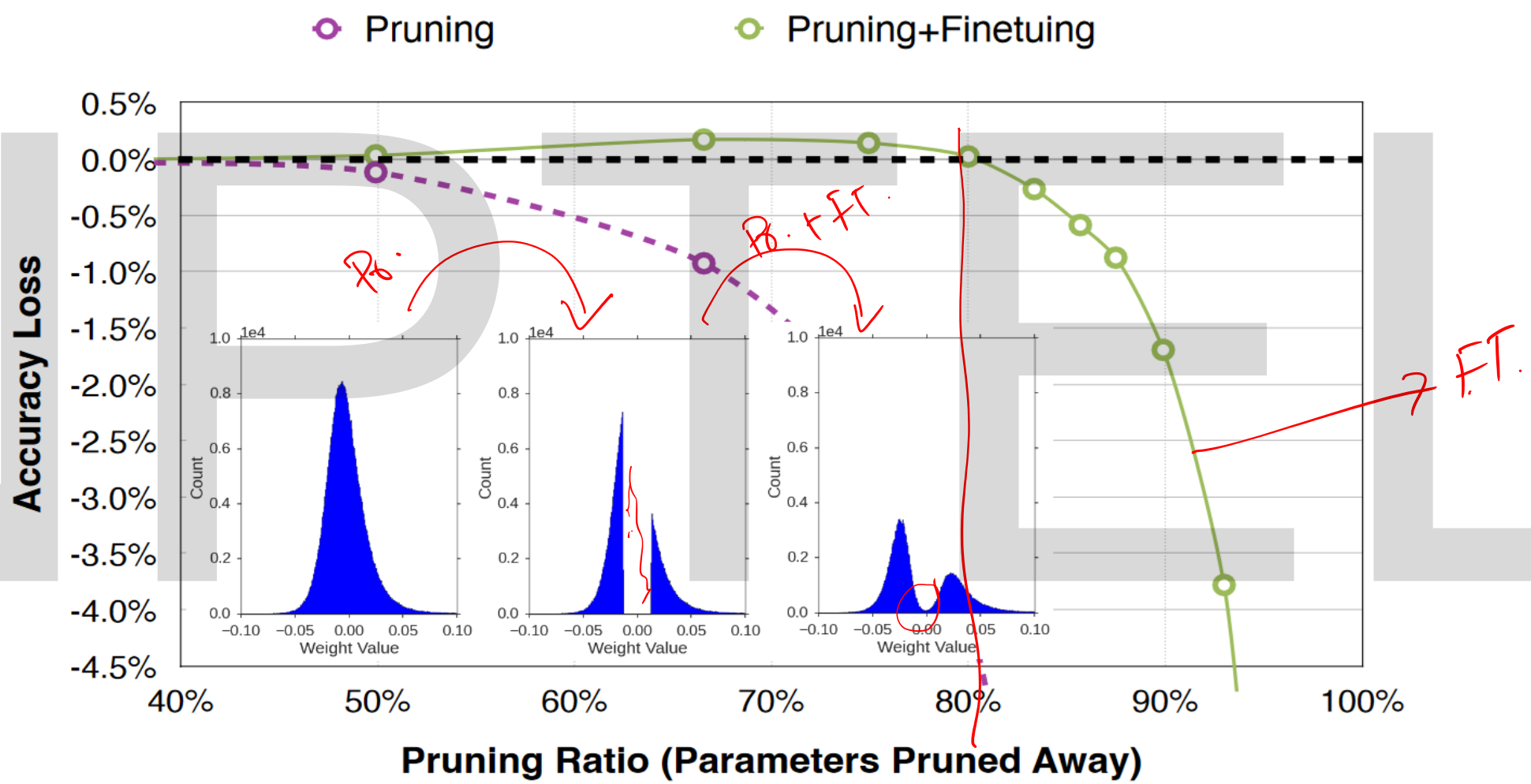
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Result of Pruning



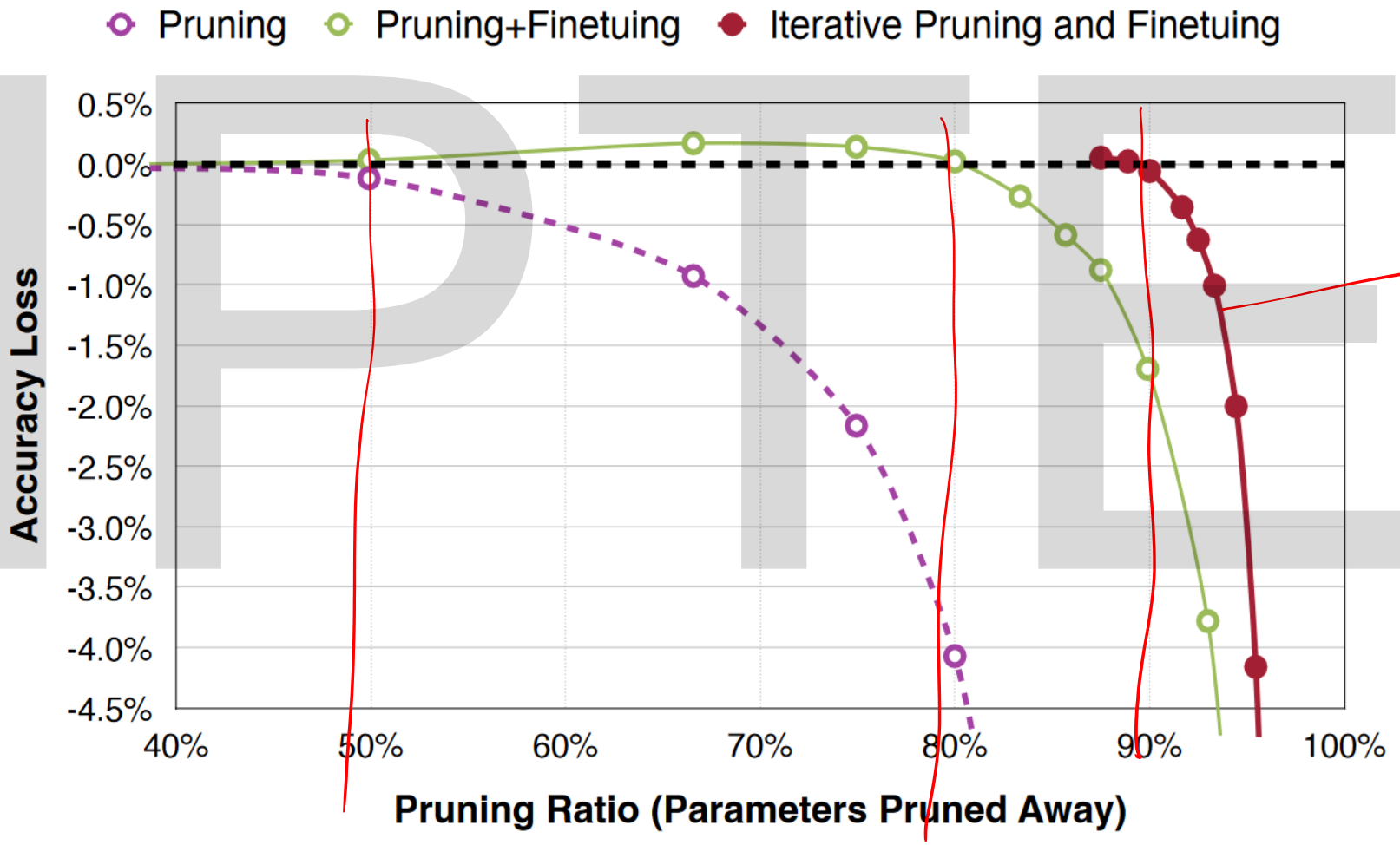
Can we recover the lost accuracy performance?

Results of Pruning and Fine tuning



Ref :Learning Both Weights and Connections for Efficient Neural Network [Han et al., NeurIPS 2015]

Iterative Pruning



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Which Parameters to Prune: Derivative-based

- Estimate the importance of a neuron by the error induced when removing that neuron
- Removing each neuron and calculating the error is computationally expensive
- Use Taylor series expansion to approximate the importance
- A perturbation δU of the parameter vector will change the objective function by

$$\delta E = \sum_i g_i \delta u_i + \frac{1}{2}\sum_i h_{ii}\delta u_i^2 + \frac{1}{2}\sum_{i \neq j} h_{ij}\delta u_i \delta u_j + O(||\delta U||^3)$$

Where, $g_i = \frac{\delta E}{\delta u_i}$ and $h_{ij} = \frac{\delta^2 E}{\delta u_i \delta u_j}$

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Which Parameters to Prune: Derivative-based

$$\delta E = \sum_i g_i \delta u_i + \frac{1}{2}\sum_i h_{ii}\delta u_i^2 + \frac{1}{2}\sum_{i \neq j} h_{ij}\delta u_i \delta u_j + O(||\delta U||^3)$$

The following assumptions can be made to simplify the approximation:

1. Diagonal: The change in error caused by deleting one filter is independent of others – cross terms are deleted
2. External: The network has already converged – the first term is deleted
3. Quadratic: Cost function is nearly quadratic – the last term is deleted

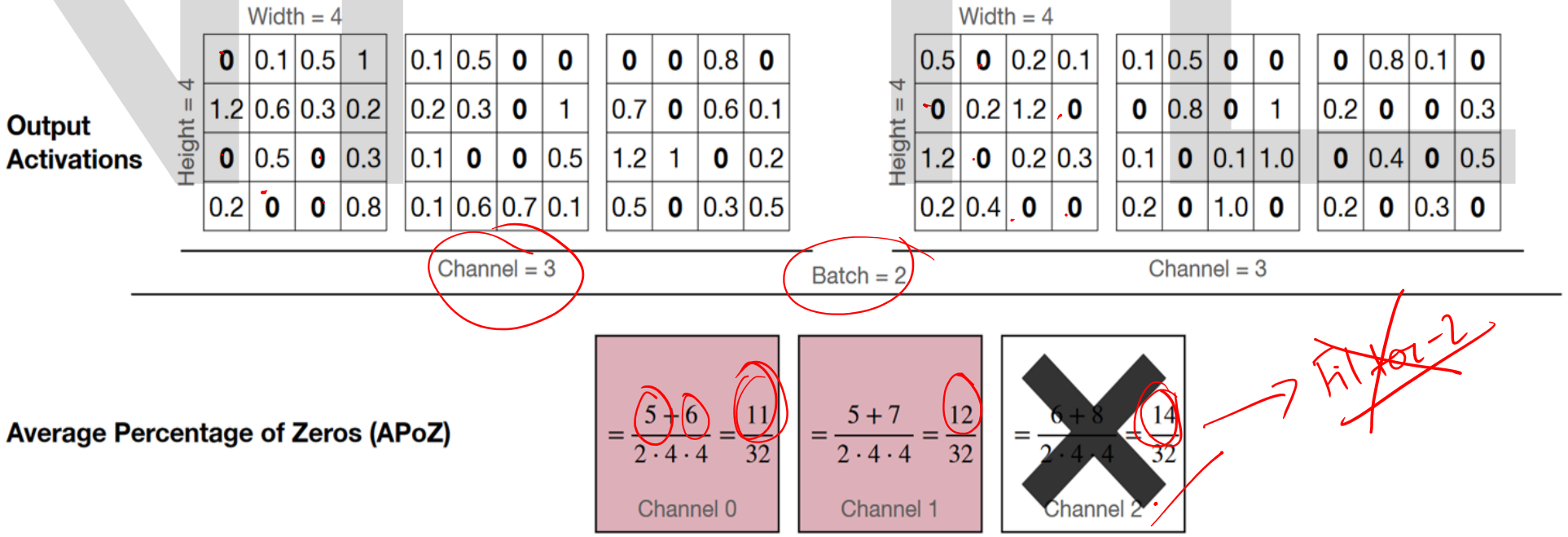
$$\delta E = \frac{1}{2}\sum_i h_{ii}\delta u_i^2$$

The parameter with the smallest $|\delta E|$ are removed.

Ref : Optimal Brain Damage, LeCun et al

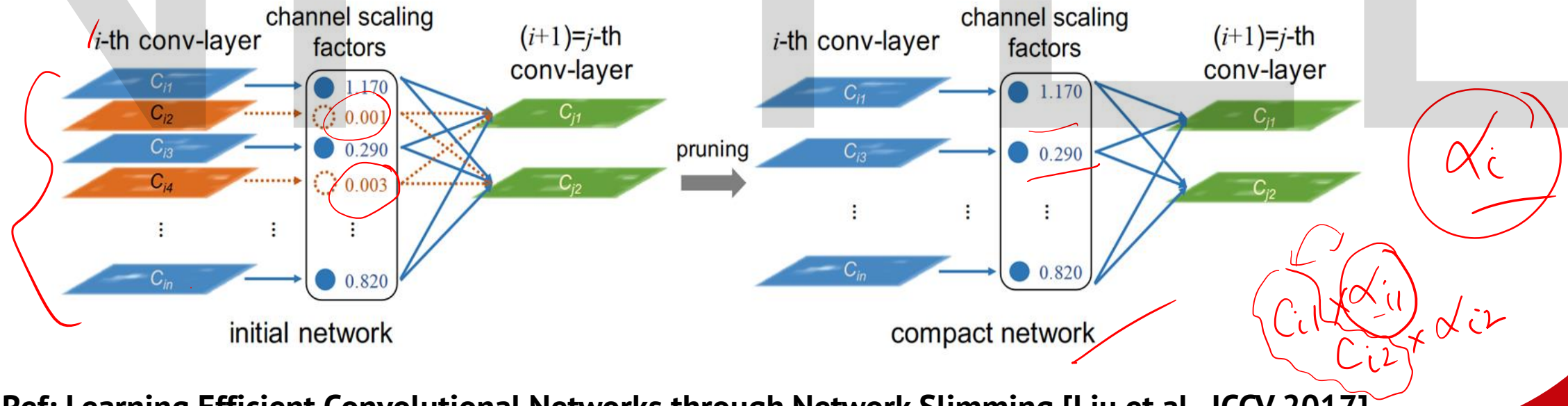
Which Parameters to Prune: Activation Based

- Zero activation neurons are redundant and can be removed without affecting the overall accuracy of the network
- Average percentage of Zero activations (APoZ) can be used to measure the importance of filters
- Filter with higher APoZ can be removed



Which Parameters to Prune: Scaling-based Pruning

- The aim of scaling-based pruning is to find the important and redundant layers while training the network itself
- For channel pruning of CNN, each channel can be associated with a scaling factor which are trainable parameters
- The channels with low scaling factors can be removed



Ref: Learning Efficient Convolutional Networks through Network Slimming [Liu et al., ICCV 2017]

The Case of Coupled Channels

- **Many modern networks architectures like ResNet make use of skip connection**
- **This skip connection contains coupled channels**
- **Channels of multiple layers are coupled and need to be pruned simultaneously**
- **Prior pruning algorithms skip pruning the coupled channels due to its complexity, but this leads to lower benefit in execution time**

Grouped Saliencies

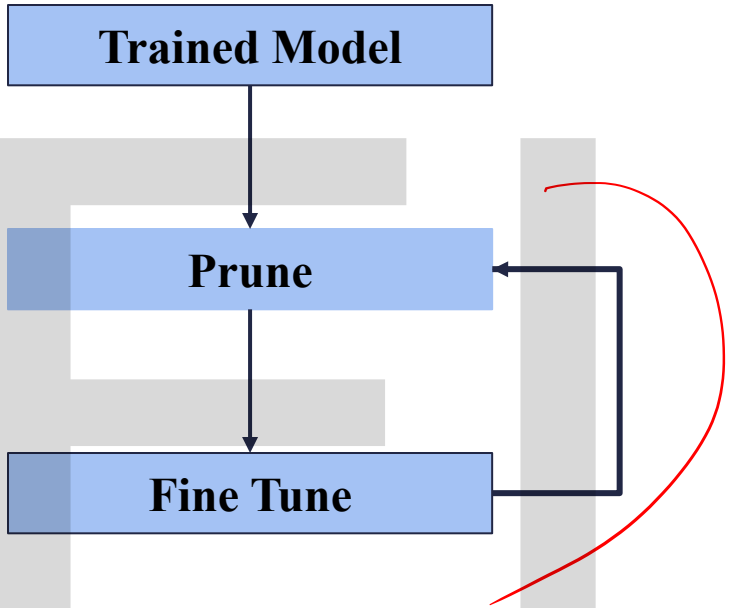
- *L .Liu et. al. PMLR 2021* proposed layer grouping algorithm to find the coupled channels. Layers in the same group shares the same pruning mask and are pruned simultaneously.

- *T Narshana et al , ICLR 2023* proposes DFC, Data Flow Coupling which abstracts the end-to-end transformation and the associated layers for an instance of coupling in a network.

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Recipe for Neural Network Pruning

- Select the pruning granularity
- Repeat
  - Select the parameters to prune
  - Remove the parameters
  - Fine tune the model



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Is this the only procedure for pruning neural network models?

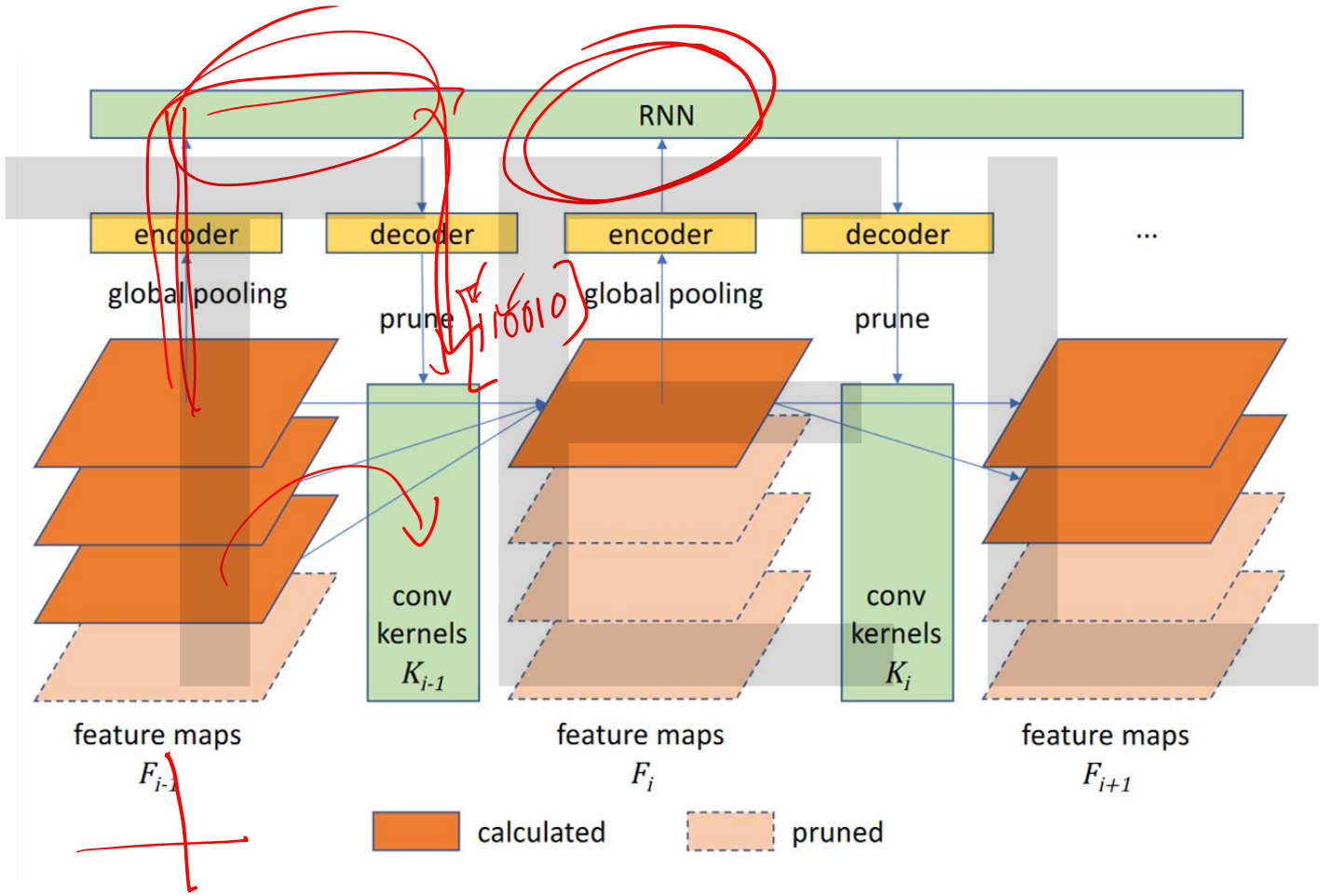
Dynamic Pruning

- **Static Pruning**
  - Static pruning permanently removes network parameters which, in general cannot be recovered
  - May lead to decrease in model capacity leading to non-recoverable accuracy
  - Static pruning doesn't take into account the current input
- **Dynamic pruning**
  - Determines in runtime which parameters: filter, connections, layers will be skipped in the forward pass
  - The network is pruned more for images which are easier to predict
  - Since the ability of the network is fully preserved, the balance point can be adjusted according to the available resources

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Dynamic Pruning (1)

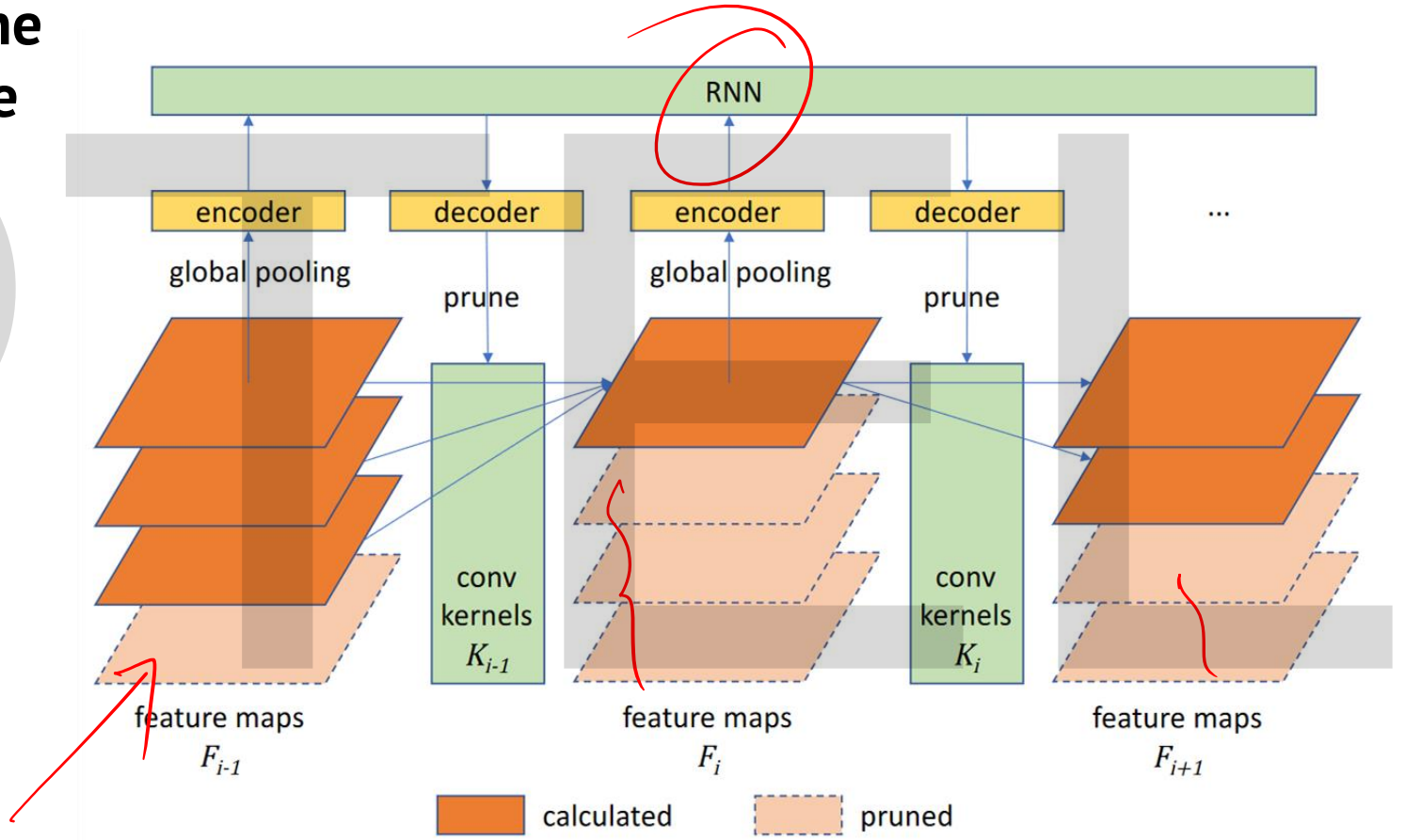
- **Dynamic Pruning or Runtime Pruning can consist of 2 sub-networks**
  - CNN Backbone
  - Decision Network
- **A mask variable can be used to denote the filters that will be used in the forward pass for the current iteration**



Ref: Pruning and Quantization for Deep Neural Network Acceleration: A Survey, Liang et al.

Dynamic Pruning (2)

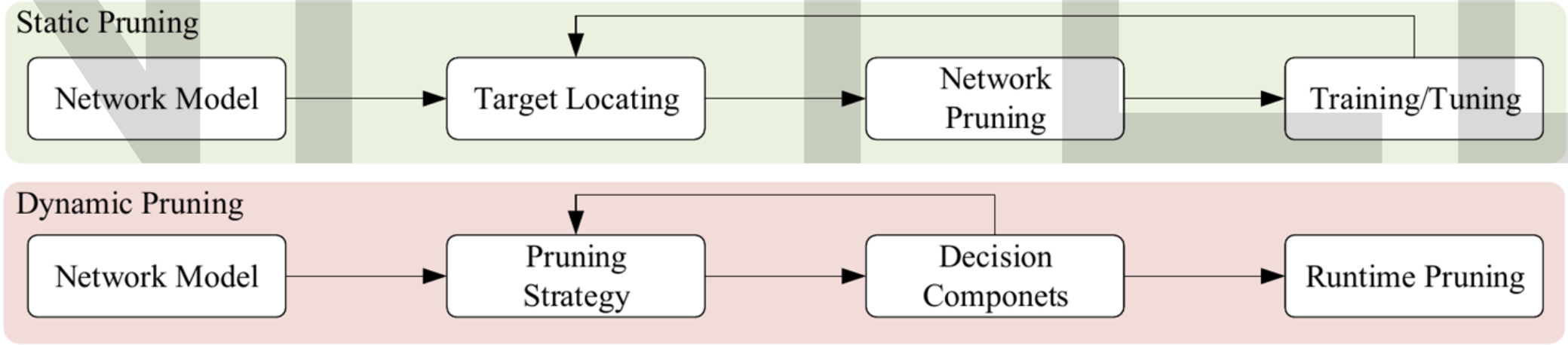
- The (i-1)th layer of the backbone network produces some feature map
- Based on this feature map, the decision network dynamically prune the kernels for the ith layer in the backbone network



Ref: Pruning and Quantization for Deep Neural Network Acceleration: A Survey, Liang et al.

Static Pruning vs Dynamic Pruning

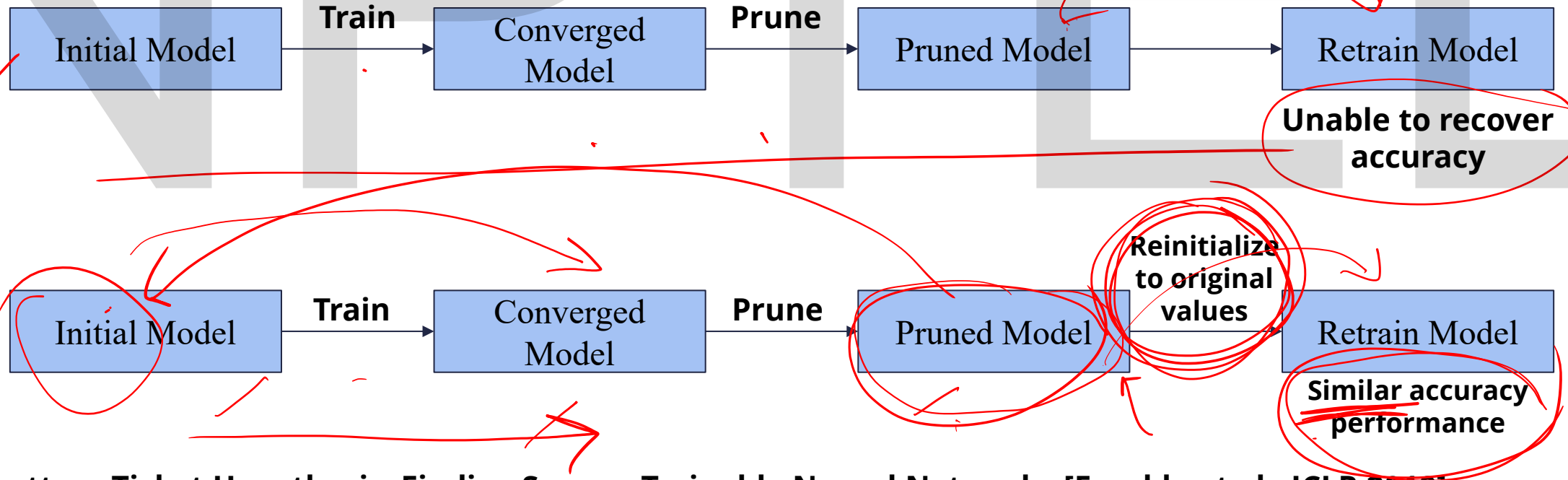
- Static pruning is performed offline prior to inference while dynamic pruning is performed at the runtime
- Dynamic pruning requires no fine tuning after pruning
- Dynamic pruning may result in higher overhead compared to static pruning



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The Lottery Ticket Hypothesis (LTH)

- **LTH asks the question whether we can train a sparse network which is a subset of some dense network, from scratch, and achieve the same level of accuracy**
- **But how do we identify the sparse subnetwork?**
  - Iterative magnitude pruning is one such way



Ref: The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks [Frankle et al., ICLR 2019]



THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)